The Librarian

Retrieval-augmented question answering over the Streamlit documentation

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1. Overview

The Librarian answers questions about **Streamlit** using only retrieved passages from its documentation (404 pages), closed GitHub issues (418) and release notes (60), split into **3,562 indexed chunks**. Every claim carries a numbered citation **[n]** that links back to its source passage, and when the passages do not contain the answer the system replies *"I don't know"* instead of guessing. It is served by a FastAPI service with a chat page, deployed on Render's free plan, with vectors stored in Qdrant Cloud.

89.7%	98.3%	100%	1.2 s	$0.00057
answer accuracy (112 Qs)	claims supported by source	"I don't know" on unanswerable	median latency	cost per question

Design principles. (1) Grounded: the generator may use nothing outside the retrieved passages. (2) Hybrid: keyword and meaning search are fused because each fails where the other succeeds. (3) Cheap: embeddings run on CPU and the vector store is on a free tier, so the only per-question cost is the language model. (4) Evolvable: a new document is one upsert, with no index rebuild.

2. Accuracy

Evaluation uses **112 questions**: 100 answerable ones with a reference answer and exact source passages, and 12 unanswerable ones about features Streamlit lacks. Retrieval and generation are scored separately; answers are graded by an independent judge (`gpt-4.1-mini`) that splits each answer into claims and checks each against the passages.

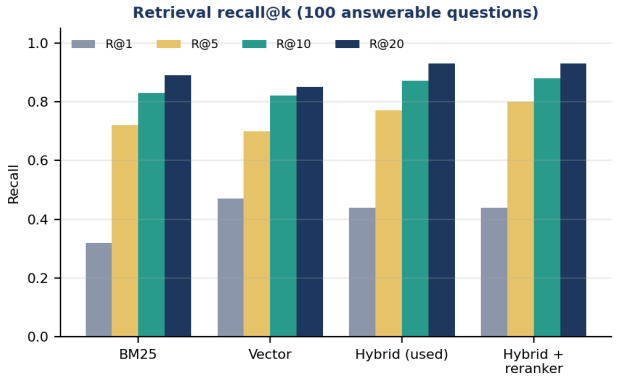


Fig. 1 - Recall@k by search mode

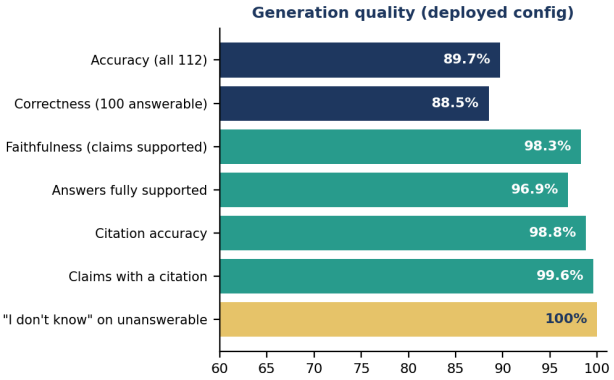


Fig. 2 - Generation quality metrics

Retrieval. Hybrid search (BM25 + vectors, fused by reciprocal rank fusion) reaches **0.87 recall@10** and 0.93 recall@20, ahead of keyword-only (0.83) and vector-only (0.82). Vector search is best at rank 1 (0.47), BM25 is best on exact error strings (recall@5 of 0.92 on error-message questions), and fusion keeps the strengths of both.

Generation. With the top 10 chunks, the system is correct on **89.7%** of all questions. Faithfulness is 98.3%, citation accuracy 98.8%, and no answer cites a nonexistent source. It refuses all 12 unanswerable questions and wrongly refuses only 4% of answerable ones.

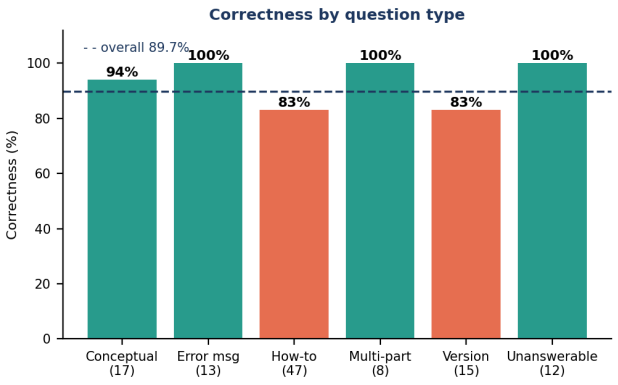


Fig. 3 - Correctness by question type

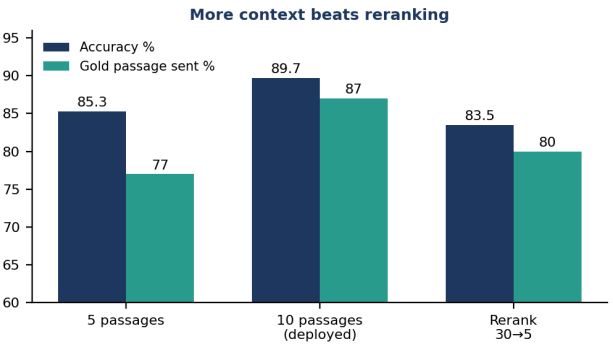


Fig. 4 - Passages sent to the model vs accuracy

Where it fails. How-to (83%) and version (83%) questions are the weak spots, and the cause is mostly retrieval: the gold passage reached the model for 87% of questions. Raising the context from 5 to 10 passages added **+4.4 points** (85.3% to 89.7%) for about $0.00025 more per question and is the deployed setting. A cross-encoder reranker improved the top ranks but **lowered** end-to-end accuracy by 1.8 points and tripled latency, so it is not used.

Configuration	Accuracy	Gold passage sent	Cost / question	p50 / p95 latency
5 passages (previous)	85.3%	77%	$0.000317	1.27 s / 2.84 s
10 passages (deployed)	**89.7% (+4.4)**	**87%**	**$0.000569**	**1.19 s / 2.45 s**
Rerank top 30 to best 5	83.5% (-1.8)	80%	$0.000289	3.07 s / 10.35 s

3. Embedding Model

Each chunk is encoded twice, and the text that is embedded is `title > section` followed by the chunk, so the heading context travels with it.

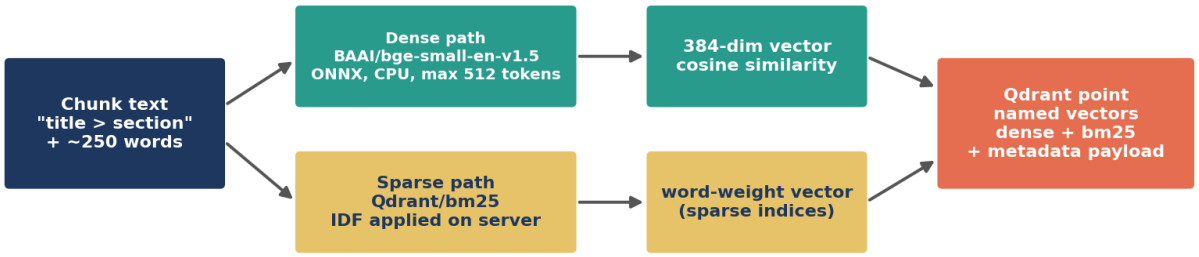


Fig. 5 - Dual encoding of every chunk into one Qdrant point

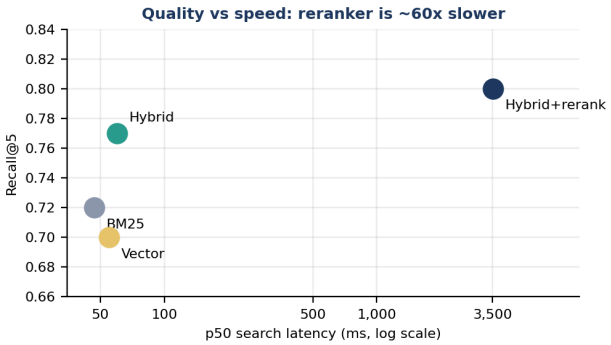


Fig. 6 - Recall@5 vs search latency (log scale)

Dense: `BAAI/bge-small-en-v1.5`, 384 dimensions, cosine similarity, run through ONNX (fastembed) on CPU.

Sparse: `Qdrant/bm25` word weights; Qdrant applies the IDF (word-rarity) part server-side.

Why this model: strong MTEB retrieval quality for its size, no GPU, no embedding API and no PyTorch, so the service stays free and fits the 512 MB host (about 300 MB for both models).

Limit: it reads at most 512 tokens, so the tail of the 17 chunks over 400 words is not embedded.

Dense and sparse are complementary. Recall@5 by question type shows it: vector search beats BM25 on conceptual (0.77 vs 0.65) and multi-part (1.00 vs 0.75) questions, while BM25 wins on error messages (0.92 vs 0.69) where exact strings matter. Hybrid fusion reaches 0.77 recall@5 overall. A cross-encoder (ms-marco-MiniLM-L-6-v2) lifts recall@3 from 0.66 to 0.75 but costs 3.5 s per query, about 60 times the hybrid search, so it was measured and not enabled.

4. Architecture

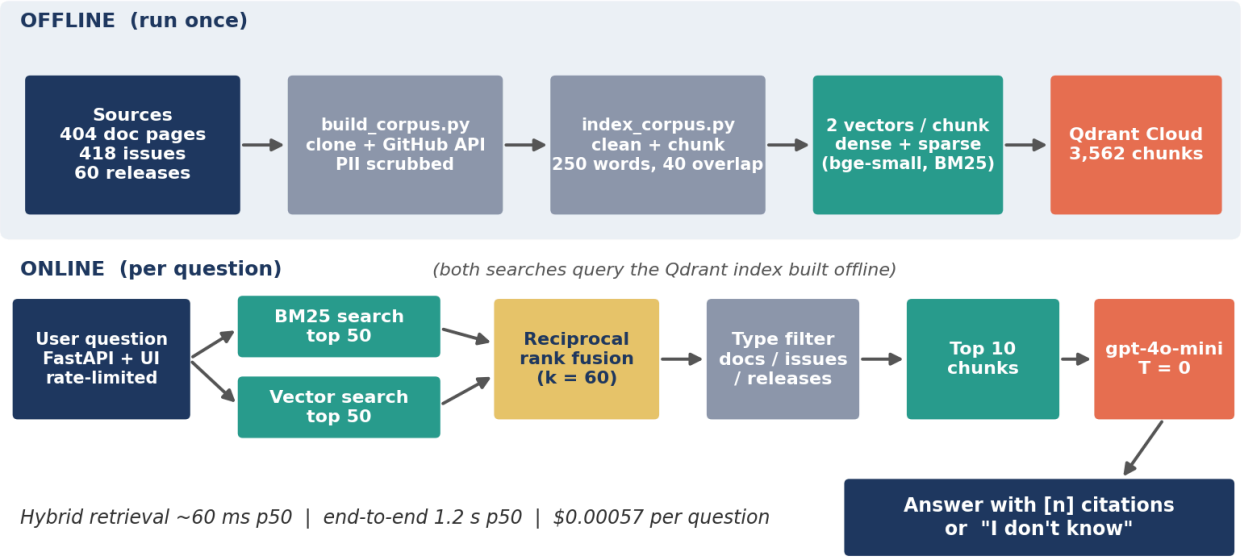


Fig. 7 - End-to-end pipeline: offline indexing (top) and online question answering (bottom)

Component	Choice	Justification
Embedding	bge-small-en-v1.5 (384-d) + BM25	Cheap, CPU-only, strong retrieval for its size
Vector database	Qdrant Cloud, one collection with named dense + bm25 vectors	Hybrid search in a single query; ids from (doc, chunk) make add/replace one upsert, no rebuild
Generator	gpt-4o-mini, temperature 0	Follows the cite-or-refuse contract at $0.15 / $0.60 per million tokens; retrieval, not the model, limits accuracy
Judge (eval only)	gpt-4.1-mini, JSON mode	Different family from the generator to avoid self-preference; re-grading moved accuracy only 86.2% to 85.3%

Operations. Chunks are about 250 words with a 40-word overlap and code blocks are never split. The service rate-limits each visitor to 10 questions per minute and the whole service to 300 per day, caches repeated questions in memory (no search, no model call), and logs cost and latency per request. 85% of the per-question cost is input tokens, so the passage count is the main cost lever.

Cost model	Per question	Per 1,000	Month @100/day	Month @10,000/day
10 passages (deployed)	**$0.00057**	**$0.57**	**$1.71**	**$170.82**
2 passages (cost floor)	$0.00015	$0.15	$0.45	$44.86

Conclusion. Sending only 2 passages cuts cost 53% but loses about 9 accuracy points, past the 2-point budget, so the deployed 10-passage setting is the best trade-off. When the right passage reaches the model it rarely strays from it, so the next gains come from better retrieval (for example, reranking only when the first search is unsure) rather than from a better prompt or a larger generator.